

Supply Chain Disruptions: Evidence from the Great East Japan Earthquake

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Research Question(s)

- **Micro-to-Macro:** Do idiosyncratic shocks to individual firms “average out” (Lucas, 1977) or propagate through the network?
- **Propagation Directions:**
 - **Downstream:** Do supplier failures starve customers of inputs?
 - **Upstream:** Do customer failures reduce demand for suppliers?
- **Network Multiplier:** How much of the 2011 GDP decline was due to direct destruction vs. supply chain “contagion”?

Economic Significance

- **Massive Scale:** Intermediate input trade accounts for over 50% of global trade and a similar share of domestic gross output.
- **The “Network Multiplier”:** If substitution elasticities are low, a small shock to a “hub” firm can have a disproportionate effect on aggregate volatility.
- **The Puzzle:** Affected prefectures = 4.6% of Japan’s output, yet GDP fell ≈ 0.43 p.p. — **four times** what direct exposure alone predicts.

The Natural Experiment: 2011 Earthquake

- **Exogeneity:** The earthquake, tsunami, and nuclear crisis were unanticipated and concentrated in the Tohoku region.
- **Directly Treated:** Firms located in the 41 designated “disaster municipalities.”
- **Indirectly Treated:** Firms *outside* the disaster area that have:
 - A supplier in the disaster area (Downstream treatment).
 - A customer in the disaster area (Upstream treatment).
- **Data:** TSR firm-level production network — 750,237 firms with supplier–customer links and annual sales panels (FY2009–FY2011).

Identification Strategy: Difference-in-Differences

The authors estimate the impact on log sales (y_{it}):

$$y_{ipst} = \gamma_i + \gamma_{pst} + \sum_{k=1}^4 \beta_k^{\text{down}} \cdot \text{Down}_i^{(k)} \times \mathbf{1}[\text{post}] + \sum_{k=1}^4 \beta_k^{\text{up}} \cdot \text{Up}_i^{(k)} \times \mathbf{1}[\text{post}] + \delta X_{i,2010} \times \mathbf{1}[\text{post}] + \varepsilon_{ipst}$$

- **Fixed Effects:** γ_i (firm) and γ_{pst} (prefecture $\times$ industry $\times$ year) absorb time-invariant heterogeneity and local/sectoral trends.
- **Geographic Distance Control:** $X_{i,2010}$ includes distance to the epicenter — isolates **network effects** from **spatial spillovers**.
- **Pre-Trends:** All 2010 coefficients are statistically insignificant — parallel trends holds before the earthquake.
- **Robustness:** (i) 60 Hz (western) firms only — rules out electricity channel; (ii) placebo network (random rewiring) — all coefficients vanish, confirming results are driven by actual supply-chain topology.

Results: Extensive and Intensive Margins

Immediate Impact (1-year post-shock)

- **Directly Hit Firms:** Sales growth fell by ≈ 3.3 percentage points relative to rest of Japan.
- **Downstream Impact:** Being a customer of a hit firm $\rightarrow$ **3.8%** decline in growth (distance 1); **2.8%** at distance 2.
- **Upstream Impact:** Being a supplier to a hit firm $\rightarrow$ **3.1%** decline in growth (distance 1); **2.1%** at distance 2.
- **Decay:** Effects propagate monotonically — significant up to distance 4 in both directions.
- **Persistence:** The shock was temporary; most firms recovered within 2 years.

Aggregate GDP Impact

The authors use a General Equilibrium (GE) model to aggregate micro-shocks:

- **Finding:** The earthquake reduced Japan's real GDP growth by **0.47 percentage points** in FY2011.
- **The Multiplier:** Counterfactual without cross-region I-O links yields only -0.21 p.p. $\Rightarrow$ network propagation accounts for **more than half** of the total impact.
- **Substitution:** Low short-run elasticity ($\hat{\sigma} < 1$) across suppliers prevents firms from “switching” quickly, magnifying the disruption.

Caveats and Limitations

- **Binary Linkages:** TSR data records link presence but not transaction values; large-volume links likely drive the results disproportionately.
- **Headquarters vs. Plant Location:** Treatment assignment uses HQ address — mitigated by single-plant robustness check, but multi-establishment firms may still introduce noise.
- **Homogeneous Links:** The model treats all supplier–customer pairs symmetrically. “Essential” bottleneck inputs (Leontief structure) could make the multiplier even larger.

Conclusion

- Production networks are a **first-order** mechanism for macroeconomic volatility.
- Shocks spread **both ways**: demand-side upstream and supply-side downstream — up to 4 links from the source.
- The network multiplier **more than doubles** the direct impact: ignoring I-O linkages dramatically understates disaster risk.
- **Final Takeaway**: Geography tells you *where* a shock lands; **network topology** determines how far it travels.